

Coursework Specification

CW_Specification_CSI_5_BDD_24/25

Read this coursework specification carefully, **based on individual submissions (via the VLE)**, it tells you how you are going to be assessed, how to submit your coursework on-time and how (and when) you'll receive your marks and feedback.

Module Code	CSI-5-BDD
Module Title	Big Data & Database Systems
Lecturer	George Ubakanma
% of Module Mark	60%
Distributed	February 2025
Submission Method	Submit online via this Module's Moodle site
Submission Deadline	18:00 - 02 Apr 2025
Release of Feedback & Marks	Feedback and provisional marks will be available in the Gradebook on Moodle within 15 working days of submission.

Coursework Aims:

- To demonstrate understanding of the module subject matter and the ability to apply it, including critical analysis and judgement. To achieve this, you are allocated to one specific case study from a set of 3 available (either case study: A, B or C). The allocation sheet is on the VLE.
- The allocated case must be analysed, and a design data model produced. Then using MS SQL Server, a relational database prototype with specific query and procedural/programming functionality must be implemented, (create your own test data using approx. 5-20 records per table); complete with a dashboard built utilising Tableau or Power BI.
- Finally, a brief 10 min video demonstration of Tasks 4(a-e) and task 5 (live execution) must be recorded using MS Teams (or similar) where you demonstrate the tasks using your prototype and explain your work.
- Discuss how you resolved technical challenges in the coding implementation. Explain how the SQL queries/procedures/triggers work or what attempts you made to make them work (if you were unsuccessful).
- During the video do a screen share so your application is full screen, but your face is also in picture (usually) in one corner of the screen narrating.
- Supply the URL to your video location in your report... Ensure your video URL works and your tutor has full access rights to the video. Any video not accessible to the tutor during marking for any reason will score zero.
- You must do the case allocated to you... failure to comply will score zero. your tutor will confirm your case allocation, if you are unsure contact your tutor.

NOTE: Any use of AI tools must be specifically declared and cited at the point(s) of use.

Coursework Details:

Type:	Report on Prototype Database Development task
Word Count:	No formal word count is required but no more than 2500 words would be expected. Footnotes will not count towards word count totals but must only be used for referencing, not for the provision of additional text. The bibliography will not count towards the word total.
Presentation:	- Any quoted material must be referenced, and a bibliography provided, however, formal referencing is not an assessed criterion – however the identification of technical resources consulted to solve specific described problems is an assessed criterion (see main task specification) - Work must be submitted as a Word document (doc/docx) - Course work must be submitted using Arial font size 11 (or larger if you need to), with a minimum of 1 line spacing - Your student number must appear at the front of the coursework. Your name must not be on your coursework.
Referencing:	Harvard Referencing should be used, see your Library Subject Guide for guides and tips on referencing.
Regulations:	Make sure you understand the University Regulations on expected academic practice and academic misconduct. Note in particular: - Your work must be your own. Markers will be attentive to both the plausibility of the sources provided as well as the consistency and approach to writing of the work. Simply, if you do the research, practical and reading, and then write it up on your own, giving reference to sources, you will approach the work in the appropriate way and will cause not give markers reason to question the authenticity of the work. - All quotations must be credited and properly referenced. Paraphrasing is still regarded as plagiarism if you fail to acknowledge the source of the ideas being expressed. TURNITIN: When you upload your work to the Moodle site it will be checked by anti-plagiarism software.

Learning Outcomes

This coursework will fully or partially assess the following learning outcomes for this module.

On completion of the module you will be able to:

- Critically discuss and evaluate a broad range of data management & big data issues including data
- administration, integrity and security.
- Discuss the continuing development of big data - database technologies/ applications and the need for continued study, reflection, and development throughout a career as a database professional.
- Design, implement, query and secure database applications and Internetworking for database systems
- Develop skills for both autonomous and independent practice, including oral and written communication skills

Assessment Criteria and Weighting

LSBU marking criteria have been developed to help tutors give you clear and helpful feedback on your work. They will be applied to your work to help you understand what you have accomplished, how any mark given was arrived at, and how you can improve your work in future.

Full details of the criteria, interpretation, weightings and rubric to be applied are included in the main task specification below.

How to get help

We will discuss this Coursework Specification in class. However, if you have related questions, please contact me:

George Ubakanma: ubakang@lsbu.ac.uk as soon as possible.

Resources

Course materials and Tutorials linked to from the exercises are expected to be the main resources required for this assignment.

Quality assurance of coursework specifications

Coursework specifications within CSI division go through internal (for new modules with 100% coursework also through external) moderation. This is to ensure high quality, consistency and appropriateness of the coursework as well as to share best practice within the CSI division.

Details of the moderators for this coursework specification are below:

Moderated (internal)	Brahim
Moderated (CSI lead)	[George Ubakanma, Feb 2025]
Signed off by (HoD/DHoD)	

-----For Internal use by CSI lead only-----

Changes required to CW?	
Examples of good practice	

*** if changes are required, moderator to complete the below:**

List of changes required	N/A
ML Response	[ML response, date]
Moderator Response	[ML response, date]

EXAMPLE CASE STUDY

London Fire Brigade (LFB).

The London Fire Brigade (LFB) is the authority responsible for running the London Fire and Rescue response also providing civil defence and emergency planning in the capital. The area protected by the authority is over 1,572 square miles and has a resident population of 9.5 million. To carry out its duties, the LFB employs some 6000 staff, of



which 5000 are operational fire fighters. Staff are either firefighters (who have a safety certification number, and an annual renewal date), admin staff (who have a job description, and a grade) or Senior officers (who have an area command number, and a 24-hour contact phone number).

Most staff are located at one of the 102 fire stations located strategically throughout the Greater London area. The Fire Stations are staffed 24 hours a day, every day of the year. The management of fire stations and a number of other associated functions are organised into a live Area Command structure. These fire stations are managed by the senior officers working in shifts with teams of fire fighters and admin staff.

The command areas are aligned with groups of London boroughs and each area incorporates a mix of buildings: city centre, commercial, industrial, residential and semi-urban development. An Assistant Chief Fire Officer commands each area. For each fire station in their area the Assistant Chief Fire Officer must ensure records are maintained on:

- All Staff having a serial number & personnel details (e.g., name, home address, home phone number), Allocated Fire Station.
- Fire Station & address/location phone no etc....

Between April 2021 and March 2023, the LFB attended nearly 217,000 incidents. Incidents are usually reported to the LFB as a result of emergency '999' telephone calls from members of the public.

These ranged from rescue of animals and helping people shut in lifts, to fires where over twenty fire appliances (fire engines or other LFB emergency rescue vehicles) are needed to control the blaze; rescuing and releasing passengers trapped or injured in the wreckage of a train crash.



The number of actual fires tackled by the LFB 33,000 over the same period with 75,000 special services and 108,000 false alarms. Resources continue to be wasted on attendance to malicious/false call incidents that are still running at a high rate. These calls waste valuable time that could be vital in the event of a genuine incident elsewhere. The control and mobilisation of appliances to incidents

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is handled by the LFB's highly trained admin staff based in the Command & Mobilising Centre at the LFB's Headquarters (HQ). The HQ computer-assisted control system opened in 2020 and mobilises the nearest available appliance to an incident, providing a swift and efficient response to emergency calls. Where possible the caller details are taken as part of the incident record.

In addition to taking '999' calls and ordering pumping appliances (fire engines) for incidents, admin staff also arrange additional back-up facilities for complex incidents such as extra breathing apparatus and other specialist equipment. They will also mobilise any appropriate special support crew (different types of firefighters) such as a petroleum officer, hazardous materials officer or a fire investigation team and monitor the availability/despatch of the LFB's fleet of 'special appliances' (e.g., turntable ladders, hydraulic platforms and specialist rescue units).



At each incident the Senior Officer (highest grade) attending will co-ordinate the actions of all crews and appliances at the location of the incident. The Senior Officer



will inform/coordinate with the HQ admin staff regarding the type of appliances that are needed to attend the incident. The senior officer will also ensure that staff attending the incident are assigned to specific duties (e.g., firefighting, rescue of casualties, taking statements from casualties & witnesses, etc...) The senior officer will also coordinate rescue operations with other emergency services e.g. Police, Ambulance, Coast

Guard.

When attending any incident, the HQ command and control centre keeps:

- A record of the Senior Officer attending the incident
- A record of which appliances are attending the incident
- A record of staff who are on board each appliance attending the incident
- Type of Incident: Fire / Rescue / False Call
- Incident location: outdoor / derelict building / residential single occupancy / residential multiple occupancy / non-residential/ commercial / industrial
- Call out Time, Arrival Time, All Clear Time, Date of incident
- Fatal Casualty caused by - gas/smoke/burns/other.
- Date of Birth of casualty. Gender of casualty. Time of Death
- Injury caused by - gas/smoke/burns/other
- Date of Birth of injured party. Gender of injured party
- Statement Made - yes/no
- Hospital Treatment required - outpatient / intensive care / casualty dept only / burns-unit

You are required to design a new prototype information system suitable for LFB's needs.

END OF CASE STUDY

CSI_5_BDD: Coursework CASE STUDY A

MARKING SCHEME AND TASKS

Complete ALL tasks listed below. Every task must be accompanied by a brief narrative (max 1 side A4 per task) discussing the task production (your attempt success/failures)

	TASK	Mark	Weight	TOTAL
1	Subject Knowledge - Design Draw a complete Entity Relationship Diagram for the case study. Include all primary/foreign keys as part of your answer. (Consider & carefully the attributes required to complete tasks 3 & 4 when you are choosing appropriate attributes)	10	0.5	5
2	Subject Knowledge - Design Produce complete Functional Dependency statements for the case study to prove that all the attributes within your entities are at least 3NF/BCNF (Consider & discuss carefully the attributes required to complete task 4 when you are choosing appropriate attributes).	10	0.5	5
3	Critical Analysis – Transforming design to implementation Using suitable relational database software (MS SQL Server) Set-up all tables, including your own, well-designed test data (minimum 20 records per table), to implement your design as a database. (Provide screenshots of all SQL creation statements including any T-SQL where appropriate, also the sample tables & data.)	10	1	10
4	Set-up and test all the following queries using Structured Query Language – SQL (provide screenshots of all SQL including any T-SQL where appropriate for each task and the task output):			
a	Write a query to find which fire station had the most genuine callouts to incidents last month?	10	0.5	5
b	Write a query to list the incidents that happened during the last month where the senior officer coordinated rescue operations with both the Police and the Ambulance service.	10	0.5	5
c	Write a query to find out which firefighters are available (not currently attending an incident) now?	10	0.5	5
d	Write a stored procedure to provide a report showing the monthly average call-out for all fire-fighters at each station?	10	0.5	5
e	Write a stored procedure to provide a report showing a complete record of incidents attended by each pumping appliance (fire engine). The procedure should be able to accept appropriate parameter values to enable dynamic search by week, month or quarter (3 months) Include appropriate attributes and totals .	10	1	10
5	Produce a dashboard combining 4-6 visualized elements, that can be used to tell an effective story about the business in your scenario. Use either Tableau or Power BI	10	1	10
6	Use MS Teams to record a video demo (10 mins max) of tasks 4 and 5. During the video you demonstrate your live execution of Task 4 in SQL Server and Task 5 in Tableau or Power BI. <i>During the video do a screen share so your application is full screen, but your face is also in picture (usually) in one corner of the screen narrating.</i> <i>Supply the URL to your video location in your report... Ensure your video URL works and the tutor has full access rights to the video.</i> <i>Any video not accessible to the tutor during marking for any reason will score zero.</i>	10	4	40
	Remember ALL tasks including screenshots evidence in a SINGLE MS Word doc for submission.			100%

CSI_5_BDD: Coursework CASE STUDY A

RUBRIC

Please note the criteria weightings and general interpretation shown in bold capitals under each criterion.

Criteria	Outstanding 100-80%	Excellent 79-70%	Very good 69-60%	Good 59-50%	Satisfactory 49-40%	Inadequate 39-30%	Very poor 29-0%
Subject Knowledge Understanding and application of subject knowledge/ application of appropriate techniques. DESIGN SUBJECT KNOWLEDGE EXPLANATION 10%	Shows sustained breadth, accuracy and detail in understanding key aspects of subject. Contributes to subject debate. Awareness of ambiguities and limitations of knowledge.	Shows breadth, accuracy and detail in understanding key aspects of subject. Contributes to subject debate. Some awareness of ambiguities and limitations of knowledge.	Accurate and extensive understanding of key aspects of subject. Evidence of coherent knowledge.	Accurate understanding of key aspects of subject. Evidence of coherent knowledge.	Understanding of key aspects of subject. Some evidence of coherent knowledge.	Some evidence of superficial understanding of subject. Inaccuracies.	Little or no evidence of understanding of subject. Inaccuracies.
Critical Analysis Analysis and interpretation of design refinement. Structuring future proofing the model CRITIQUE DESIGN TRANSFER TO CODE 10%	Outstanding demonstration of critical analysis of the possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Excellent demonstration of critical analysis of the possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Very good demonstration of critical analysis of the possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Good demonstration of critical analysis of the possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Demonstration of critical analysis of the key possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Trivial demonstration of critical analysis of the possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Little or no critical analysis has been demonstrated.
Testing and Problem-Solving Skills Design, implementation, testing and analysis of product/process/ system/idea/solution(s) to practical questions/ problems IMPLEMENTATION OF SQL + PROCEDURES 30%	Outstanding implementation of all required software, with near perfectly organised, formatted and documented source code, and documented demonstration of runtime behaviour.	Excellent implementation of all required software, with well organised, formatted and documented source code provided	Competent implementation of all required software, with well organised, formatted and documented source code, and documented demonstration of runtime behaviour.	Implementation of all required software, with well organised, formatted and documented source code, and documented demonstration of runtime behaviour, with some missing/incorrect functionality or poor quality.	Implementation of most of the required software, with well organised, formatted and documented source code, and documented demonstration of runtime behaviour, with some missing/incorrect functionality or poor quality.	Implementation of only part of the required software, with well organised, formatted and documented source code, and documented demonstration of runtime behaviour, with some missing/incorrect functionality or poor quality.	Little or no functionality has been implemented.
Practical Competence Skills to apply theory to practice or to test theory USE OF DASHBOARD TOOLS 10%	Outstanding descriptions of factual information, programming techniques or theoretical explanations being found in technical or theoretical reference material.	Excellent explicit descriptions of all factual information, programming techniques or theoretical explanations that were found in technical or theoretical reference material.	Good explicit descriptions of all factual information, programming techniques or theoretical explanations that were found in technical or theoretical reference material.	Reasonable descriptions of most factual information, programming techniques or theoretical explanations that were found in technical or theoretical reference material.	Basic examples of the main factual information, programming techniques or theoretical explanations that were found in technical or theoretical reference material.	Some trivial examples of factual information, programming techniques or theoretical explanations being found in technical or theoretical reference material.	Little or no evidence of factual information, programming techniques or theoretical explanations being found in technical or theoretical reference material.
Personal and Professional Development Evidence Management of learning through self-direction, planning and reflection VIDEO DEMO 40%	Outstanding organisation, structure, presentation, narrative voice and language.	Excellent organisation, structure, presentation, narrative voice and language.	Very good organisation, structure, presentation, narrative voice and language.	Good organisation, structure, presentation, narrative voice and language.	Satisfactory organisation, structure, presentation, narrative voice and language.	Poor organisation, structure, presentation, narrative voice and language.	Does not constitute a serious attempt at the assignment.

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EXAMPLE CASE STUDY

LSBU Manufacturing Ltd

LSBU owns a factory manufacturing various spare parts for commercial vehicles (e.g., vans, heavy goods vehicles, coaches, buses, etc...). They want to implement a database system to manage their operation.



Recently a team of analysts worked within the company and identified some basic requirements that LSBU recognises as specifications for its potential new information system, as follows:

The factory has several departments. (e.g., Design / Production/ Maintenance/Sales etc.) A department may have many employees, although each department must have at least three employees. Every employee works for one and only one department. Every department has a manager - with only one manager per department. Clearly a manager is an employee of the company, but not all employees are managers.

A department may have many production machines, and every machine is assigned to a specific department. The machines are of varying types: Manual Lathes / Computer Aided Manufacturing (CAM) cutters /Machine presses etc... Each machine has a purchase date, and a record is also kept about its service history.



The machines are used to produce engineering products which LSBU's sales team then sell on to commercial clients. Within LSBU the spare parts it manufactures are referred to as products. Every product has a product number, description, manufacture date/time, manufacturing cost and base sales price... which is calculated as the (manufacturing cost + 45%)

Due to the very high market demand for its engineering products for commercial vehicles, LSBU likes to keep each machine as busy as possible 24hours-per day, 7 days per week, working on the manufacture of its range of products.



Most of its machines are programmable and utilise Computer Aided Design - Computer Aided Manufacturing (CAD-CAM) interfaces to take product design instructions directly from Designer computer

CSI_5_BDD: Coursework CASE STUDY B

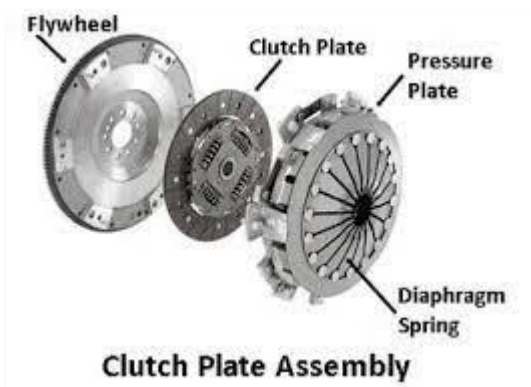
workstations and have the capability to manufacture the products to the required specifications automatically, sometimes without human intervention.

Other machines are manual i.e., they require an Operator (a skilled employee) who is responsible for manually operating the machine(s). Each operator belongs to a trade union and has specific qualifications certified technical skills associated with their job including, health and safety at work certifications which all operators must have renewed every 2 years.

Due to a system of working multiple shifts, several operators operate the same machine at different times of the day. All operators are routinely assigned to work on only one machine and no operator is kept idle. A machine is never kept idle either, except when it is out for maintenance.



For each product that a machine produces the product is also given a unique serial number (this is in addition to the existing product id) so that products with the same id e.g., every flywheel is product id 500124, but each actual flywheel produced will also have a unique serial number e.g., 0001, 0002, 0003 etc...



Because of the busy work schedule and long running hours, a machine may be reported for maintenance numerous times. Regular maintenance is actually done on a machine every 72 hours.

If a machine is reported as being faulty then a special maintenance call can be issued once on a given day. Some machines are so new they may not have required any maintenance yet.

The products that are produced on the machines vary widely in nature (e.g., plates/ gears/ bearings/nuts/bolts/screws etc..). A product can be an assembly of several different component products (e.g., an engine or gearbox/clutch may have many products as its parts.) or a product can be a single piece.

Every product and component go through one or more machines for appropriate production processes. Likewise, several products may go through a particular machine for some standard safety testing process.



You are required to design a new prototype information system suitable for LSBU Manufacturing's needs.

END OF CASE STUDY

CSI_5_BDD: Coursework CASE STUDY B

MARKING SCHEME AND TASKS

Complete ALL tasks listed below. Every task must be accompanied by a brief narrative (max 1 side A4 per task) discussing the task production (your attempt success/failures)

	TASK	Mark	Weight	TOTAL
1	Subject Knowledge - Design Draw a complete Entity Relationship Diagram for the case study. Include all primary/foreign keys as part of your answer. (Consider & discuss carefully the attributes required to complete tasks 3 & 4 when you are choosing appropriate attributes)	10	0.5	5
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4	Set-up and test all of the following queries using Structured Query Language – SQL <i>(provide screenshots of all SQL including any T-SQL where appropriate for each task and the task output):</i>			
a	Write a query to find the operators who have worked the highest and lowest number of shifts in the last month.	10	0.5	5
b	Write a query to find all machines that have been used to produce any of the following parts (Flywheel, Clutch plate, Pressure plate) in the past 3 days. In the output list each machine along with a list of all parts it has produced and the production date, then the next machine and its parts produced with production date, etc...	10	0.5	5
c	Write a query to produce a list of all machines that have been reported for maintenance issues on more than 3 occasions.	10	0.5	5
d	Write a procedure to demonstrate the business rule that an employee's pay may not exceed their manager's pay.	10	0.5	5
e	Write a procedure to provide a monthly report showing the total salary bill for each department. The output should include individual employee details for every company employee and a calculated total for each department. The procedure should be able to accept appropriate parameter values to enable dynamic search by week, month or quarter (3 months) Include appropriate attributes and totals in your report	10	1	10
5	Produce a dashboard combining 4-6 visualized elements, that can be used to tell an effective story about the business in your scenario. Use either Tableau or Power BI	10	1	10
6	Use MS Teams to record a video demo (10 mins max) of tasks 4 and 5. During the video you demonstrate your live execution of Task 4 in SQL Server and Task 5 in Tableau or Power BI. <i>During the video do a screen share so your application is full screen, but your face is also in picture (usually) in one corner of the screen narrating. Supply the URL to your video location in your report... Ensure your video URL works and the tutor has full access rights to the video. Any video not accessible to the tutor during marking for any reason will score zero.</i>	10	4	40
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EXAMPLE CASE STUDY

LSBU SMART HOME

LSBU runs a small company (Smart Home) that designs, installs and maintains smart home technologies using the latest Internet of Things (IoT) sensors (e.g. for heat/humidity/smoke, door-operation/motion detection, security applications, home automation and device monitoring, etc...) for its clients.



Each client may own several buildings, and each building may belong to multiple clients e.g. multiple flats or apartments within the same building.

LSBU offers a design service to its customers where it plans the layout and location of its IoT sensors and other specialist feature devices that it can incorporate into a building.

Each design is unique and is not duplicated. Not every building owned by a client is totally of LSBU's own design, as clients have the right to use rival firms to perform their IoT sensor

installs if they choose.

For example LSBU may do the design only and a 3rd party contractor may be employed (at the customer's request) to do the installation. Each design uses several types of IoT sensors, and may also include specialist equipment such as HD-CCTV, ultrasonic motion detection systems etc... Both the IoT sensors and the other specialist equipment used may appear in many different LSBU designs.

A feature that LSBU would like to introduce to its designs is full remote control of the smart building via the customer's mobile phone or other handheld device. They are in discussions with several software development companies to build a mobile app that can control all devices connected within a specific customer's smart home installation.

LSBU keeps a record of which IoT sensors and controller units complement each other ensuring the best integration and compatibility between devices in each design-installation.

In order to ensure that each type of IoT sensor & controller device required is available when needed, LSBU ensures that it can obtain stocks of each IoT sensor from a number of suppliers

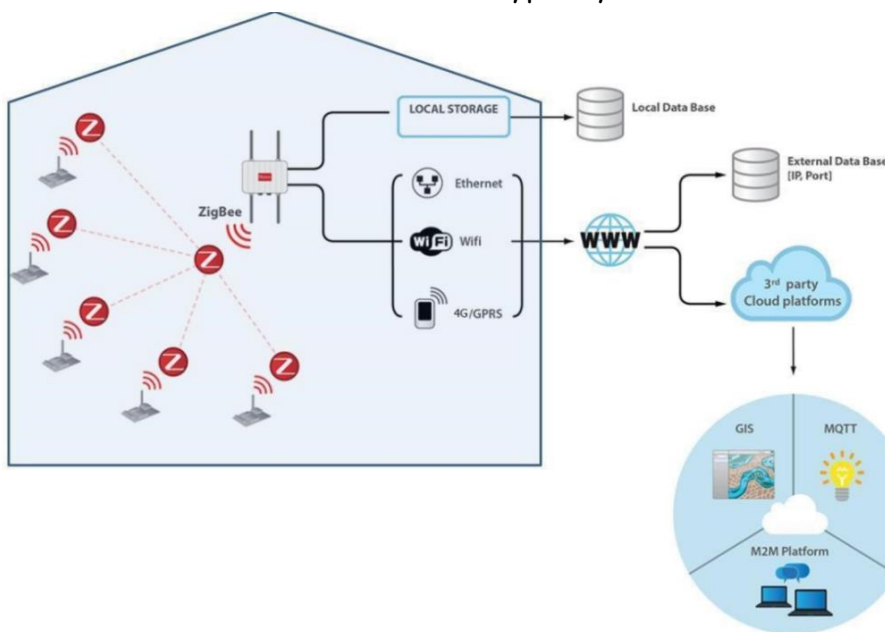


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(e.g. Libelium, Shenzhen etc...). Each supplier carries a large stock of various IoT sensors, controllers and other equipment. LSBU makes regular checks online to ensure stock availability but would like to automate this process.

LSBU assigns its staff on a team basis to work on installing each design. Each team consists of at least two members of staff. A record of cameras, sensors, controllers used on each job and the specific staff installing each component is kept. Also the staff contact details (name, specialist expertise, mobile phone number, etc...) are kept in order to ensure that staff can be re-assigned at the last minute to cope with unforeseen emergencies. Behind all of this technology, databases are being developed (your advice is also being sought for this) to monitor/report and provide analytics on the performance of each smart home and the various sensors that it contains.

Each IoT sensor typically sends regular readings (wirelessly) to a controller device (located within the home). The controller device typically co-ordinates communications between the sensors



connected to it using a protocol such as ZigBee. The controller will record the individual smart home's IoT sensor data locally on a database for direct storage and analysis.

Periodically (either daily or weekly) the controller will also make a secure encrypted wireless broadcast of its' latest same smart home data via the web to LSBU's data centre, where a central copy of all of LBSU's clients smart

home data is stored in database which is only accessible remotely via a secured IP Port address. LSBU would like to make use of 3rd party cloud-based platforms solutions to enhance its offering, but needs to develop more expertise in this area, and is happy to take well researched advice.

LSBU currently produces invoices once a month for its designs and installations. LSBU bill separately for maintenance callouts. LSBU only accept a single payment by credit/debit card for each invoice, within 28 days of the invoice date.

They would like to investigate the possibility to incorporate new payment mechanisms e.g. PayPal, and innovative mobile payment solutions e.g. Google Pay. LSBU would like to know whether it is employing technology to best effect. It is looking to gain efficiencies and improve its database functionality in all areas.

You are required to design a new prototype information system suitable for LSBU Smart Home's needs.

END OF CASE STUDY

CSI_5_BDD: Coursework CASE STUDY C

MARKING SCHEME AND TASKS Complete ALL tasks listed below. Every task must be accompanied by a brief narrative (max 1 side A4 per task) discussing the task production (your attempt success/failures)

	TASK	Mark	Weight	TOTAL
1	Subject Knowledge - Design Draw a complete Entity Relationship Diagram for the case study. Include all primary/foreign keys as part of your answer. (Consider & discuss carefully the attributes required to complete tasks 3 & 4 when you are choosing appropriate attributes)	10	0.5	5
2	Subject Knowledge - Design Produce complete Functional Dependency statements for the case study to prove that all of the attributes within your entities are in at least 3NF/BCNF (Consider & discuss carefully the attributes required to complete task 4 when you are choosing appropriate attributes).	10	0.5	5
3	Critical Analysis – Transforming design to implementation Using suitable relational database software (MS SQL Server) Set-up all tables, including your own, well-designed test data (minimum 20 records per table), to implement your design as a database. <i>(Provide screenshots of all SQL creation statements including any T-SQL where appropriate, also the sample tables & data.)</i>	10	1	10
4	Set-up and test all of the following queries using Structured Query Language – SQL <i>(provide screenshots of all SQL including any T-SQL where appropriate for each task and the task output):</i>			
a	Write a query to show details of clients, their property location(s) and the total value of Smart Home designs installed, specifically for those clients who have made the most expensive and the least expensive total values.	10	0.5	5
b	Write a query to calculate individual totals for the number of complete, incomplete and cancelled WiFi product orders from each of its suppliers.	10	0.5	5
c	Write a query to find the details of specialist staff who are available for booking onto installation jobs this week.	10	0.5	5
d	Implement using triggers and/or stored procedures automatic prevention of double-booking of specialist staff during the installation booking process.	10	0.5	5
e	Write a Stored Procedure that can generate a complete costed device list and total cost for any installation (s) in a specified time related period. The output must also include client and device details. The procedure should be able to accept appropriate parameter values to enable dynamic search by week, month or quarter (3 months) Include appropriate attributes and totals in your report	10	1	10
5	Produce a dashboard combining 4-6 visualized elements, that can be used to tell an effective story about the business in your scenario. Use either Tableau or Power BI	10	1	10
6	Use MS Teams to record a video demo (10 mins max) of tasks 4 and 5. During the video you demonstrate your live execution of Task 4 in SQL Server and Task 5 in Tableau or Power BI. <i>During the video do a screen share so your application is full screen, but your face is also in picture (usually) in one corner of the screen narrating. Supply the URL to your video location in your report... Ensure your video URL works and the tutor has full access rights to the video. Any video not accessible to the tutor during marking for any reason will score zero.</i>	10	4	40
	Remember ALL tasks including screenshots + all code evidence in a SINGLE MS Word doc for submission.			100%

CSI_5_BDD: Coursework CASE STUDY C

RUBRIC Please note the criteria weightings and general interpretation shown in bold capitals under each criterion.

Criteria	Outstanding 100-80%	Excellent 79-70%	Very good 69-60%	Good 59-50%	Satisfactory 49-40%	Inadequate 39-30%	Very poor 29-0%
Subject Knowledge Understanding and application of subject knowledge/ application of appropriate techniques. DESIGN SUBJECT KNOWLEDGE EXPLANATION 10%	Shows sustained breadth, accuracy and detail in understanding key aspects of subject. Contributes to subject debate. Awareness of ambiguities and limitations of knowledge.	Shows breadth, accuracy and detail in understanding key aspects of subject. Contributes to subject debate. Some awareness of ambiguities and limitations of knowledge.	Accurate and extensive understanding of key aspects of subject. Evidence of coherent knowledge.	Accurate understanding of key aspects of subject. Evidence of coherent knowledge.	Understanding of key aspects of subject. Some evidence of coherent knowledge.	Some evidence of superficial understanding of subject. Inaccuracies.	Little or no evidence of understanding of subject. Inaccuracies.
Critical Analysis Analysis and interpretation of design refinement. Structuring future proofing the model CRITIQUE DESIGN TRANSFER TO CODE 10%	Outstanding demonstration of critical analysis of the possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Excellent demonstration of critical analysis of the possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Very good demonstration of critical analysis of the possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Good demonstration of critical analysis of the possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Demonstration of critical analysis of the key possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Trivial demonstration of critical analysis of the possible design strategies that could be used to meet the software requirements, and evaluation of the approaches chosen.	Little or no critical analysis has been demonstrated.
Testing and Problem-Solving Skills Design, implementation, testing and analysis of product/process/ system/idea/solution(s) to practical questions/ problems IMPLEMENTATION OF SQL + PROCEDURES 30%	Outstanding implementation of all required software, with near perfectly organised, formatted and documented source code, and documented demonstration of runtime behaviour.	Excellent implementation of all required software, with well organised, formatted and documented source code provided	Competent implementation of all required software, with well organised, formatted and documented source code, and documented demonstration of runtime behaviour.	Implementation of all required software, with well organised, formatted and documented source code, and documented demonstration of runtime behaviour, with some missing/incorrect functionality or poor quality.	Implementation of most of the required software, with well organised, formatted and documented source code, and documented demonstration of runtime behaviour, with some missing/incorrect functionality or poor quality.	Implementation of only part of the required software, with well organised, formatted and documented source code, and documented demonstration of runtime behaviour, with some missing/incorrect functionality or poor quality.	Little or no functionality has been implemented.
Practical Competence Skills to apply theory to practice or to test theory. USE OF DASHBOARD TOOLS 10%	Outstanding descriptions of factual information, programming techniques or theoretical explanations being found in technical or theoretical reference material.	Excellent explicit descriptions of all factual information, programming techniques or theoretical explanations that were found in technical or theoretical reference material.	Good explicit descriptions of all factual information, programming techniques or theoretical explanations that were found in technical or theoretical reference material.	Reasonable descriptions of most factual information, programming techniques or theoretical explanations that were found in technical or theoretical reference material.	Basic examples of the main factual information, programming techniques or theoretical explanations that were found in technical or theoretical reference material.	Some trivial examples of factual information, programming techniques or theoretical explanations being found in technical or theoretical reference material.	Little or no evidence of factual information, programming techniques or theoretical explanations being found in technical or theoretical reference material.
Personal and Professional Development Evidence Management of learning through self-direction, planning and reflection. VIDEO DEMO 40%	Outstanding organisation, structure, presentation, narrative voice and language.	Excellent organisation, structure, presentation, narrative voice and language.	Very good organisation, structure, presentation, narrative voice and language.	Good organisation, structure, presentation, narrative voice and language.	Satisfactory organisation, structure, presentation, narrative voice and language.	Poor organisation, structure, presentation, narrative voice and language.	Does not constitute a serious attempt at the assignment.